

Chem-E Car Propulsion Mechanism: Thermal Electric Generators



Gustavo De la Torre

California State University Long Beach

Chemical Engineering Department

CHE 450 Laboratory II

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Abstract

This report details the design and development of a chemically powered vehicle for the Chem-E-Car Competition. The propulsion system utilized thermoelectric generators (TEGs) to convert thermal energy from an acid-base reaction and the cooling effect of dry ice into electrical power. Performance was optimized by varying TEG size, quantity, and chemical reaction parameters. The final design featured a 3D-printed carbon fiber container with two hot boxes and integrated heat sinks. This innovative system enabled the success of the team during the competition.

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Introduction

For the Chem-E-Car Competition, a vehicle propulsion system was developed using thermoelectric generators (TEGs) to convert thermal energy into electrical power. The system employed an exothermic acid-base reaction (HCl and NaOH) as the heat source and dry ice as the cold sink. The experimental design focused on optimizing system performance through a systematic variation of parameters, including TEG sizes and quantities, alternative chemical reactions, and thermal management strategies.

The optimized system featured a 3D-printed carbon fiber container housing the hot and cold reservoirs, with TEGs connected in series to maximize voltage output. Testing and analysis revealed that TEGs without copper plates yielded the best voltage output, and the carbon fiber container provided adequate durability in the acidic environment. The final design, utilizing 8 TEGs with the HCl/NaOH reaction and 250g-350g of dry ice, successfully powered a 12V DC motor, demonstrating the feasibility of this approach.

System performance analysis showed an initial peak output of 24V and 1.3A, which declined to 3V and 0.204A over 10 minutes. This indicates that while the design effectively converts chemical energy into electrical power, it is best suited for short-duration applications. Future research may focus on enhancing sustained power generation and incorporating voltage sensors and capacitors for improved performance and control.

Experimental Design and Apparatus

The experimental design centered on harnessing the Seebeck effect using thermoelectric generators (TEGs) to convert a temperature differential into electrical energy for vehicle propulsion. The core setup involved creating a significant temperature gradient by utilizing an exothermic acid-base reaction as the hot side and dry ice as the cold side. Multiple TEGs were arranged to maximize the voltage and current output from this thermal gradient, with the generated electricity intended to power a 12V DC motor for the car. Various configurations and materials were tested throughout the project to optimize the efficiency and power output of this thermoelectric system.

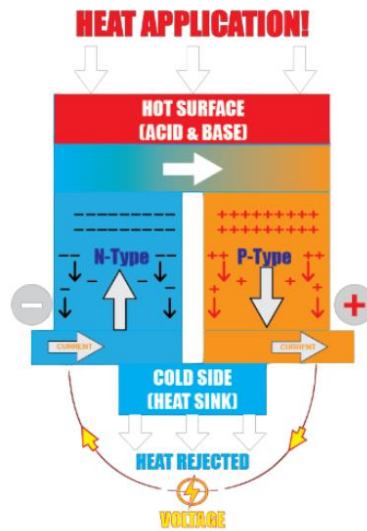


Image 1 shows a simplified version of the Seebeck effect

The apparatus primarily consisted of components housed within custom-designed 3D-printed carbon fiber containers. These containers were specifically fabricated to hold the hot side reaction mixture and the dry ice-cold sink while facilitating efficient heat transfer to the TEGs. The TEGs were strategically placed between the hot and cold sections, bonded with epoxy and

thermal paste to ensure good thermal contact. The system also incorporated heat sinks on the cold side to enhance cooling from the dry ice and a pump to circulate the hot side solution, ensuring a well-mixed reaction and consistent heat distribution. A manual switch was included for system control.

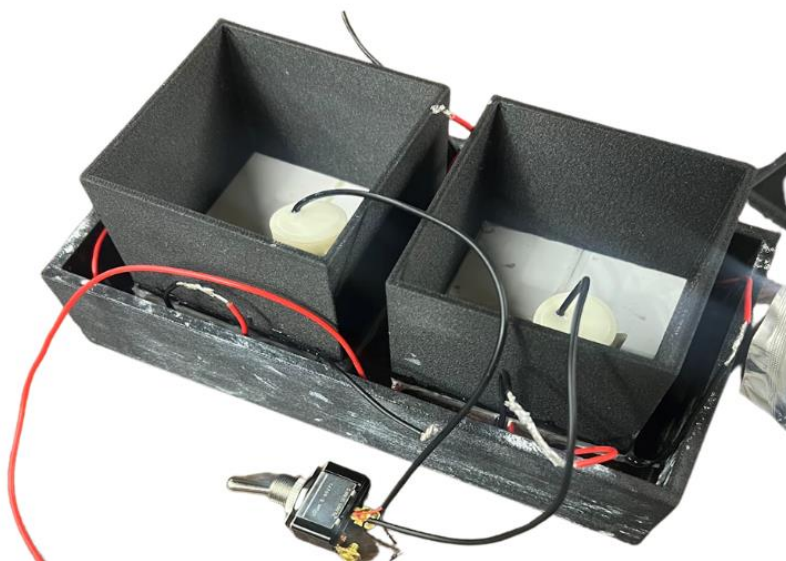


Image 2 shows the inside of the hot boxes

The experimental procedure involved several key steps. First, the chemical reactants, 3M HCl and 10 grams of solid NaOH in each container, were prepared following strict safety protocols, including dilution of the concentrated acid while wearing the proper PPE. The TEGs were wired in series (four per hot box, totaling eight in the final design) and tested to ensure proper electrical connections. The 3D-printed containers were assembled with the TEGs and the heat sinks. Experiments were conducted by initiating the acid-base reaction in the hot box and adding dry ice to the cold box, then measuring the voltage and current output over time using a multimeter or to evaluate the system's performance.

Procedure

This document provides a detailed outline for the Procedure section of your lab report, drawing from the information available in the provided files.

Procedure Outline

1. Acid-Base Reaction Preparation

- Prepare 330 ml of 3M HCl solution by diluting 12M HCl stock solution in a fume hood.
- Wear proper PPE at all times: gloves, acid-resistant apron, goggles, and face shield.
- Measure 10 g of NaOH and set it in each box.
- Gradually add the HCl to the NaOH using the pumps when switch is on.

2. TEG Wiring and Testing

- Connect the thermoelectric generators (TEGs) in series. The final design used four TEGs per hot box, totaling eight TEGs.
- Use insulated wiring for all connections.
- Check all electrical connections thoroughly before powering the system or proceeding with the experiment.

3. System Assembly and Preparation for Experiment

- Assemble the 3D-printed carbon fiber containers, including the hot box for the reaction and the cold box for the dry ice.

- Securely attach the TEGs to the hot box using epoxy, ensuring good thermal contact.
- Apply thermal paste evenly to the cold side of the TEGs to facilitate efficient heat transfer to the heat sinks.
- Place the heat sinks in the cold box, ensuring they are in direct contact with the dry ice.
- Integrate the pump for circulating the hot side solution and the manual switch for system control.
- Add the appropriate amount of dry ice (optimal range was 250g–350g) to the cold box just prior to starting the experiment.

4. Experimental Run and Data Collection

- Initiate the acid-base reaction by combining the prepared HCl and NaOH in the hot box.
- Immediately begin monitoring and recording the voltage and current output generated by the TEG system over time using a multimeter or oscilloscope.
- Collect data points at regular intervals (e.g., every minute) for a specified duration (e.g., 10 minutes) to observe the system's performance as the reaction proceeds and the temperature gradient changes.
- Record any qualitative observations about the reaction, temperature changes, or system behavior during the experimental run.

- Follow proper chemical waste neutralization and disposal protocols for the reacted chemicals after the experiment is complete.

Results and Discussion

The primary data presented in Figure 1 illustrates the voltage and current output of the thermoelectric generator (TEG) system over a 10-minute period. The data, recorded as voltage (V) and current (A) at 1-minute intervals, shows a clear decreasing trend for both parameters. Initially, the system exhibits a high output of 24V and 1.3A, which indicates a substantial electrical energy generation at the onset. However, as time progresses, both voltage and current diminish, with the final measurements at the 10-minute mark dropping to 3V and 0.204A. This decline suggests a reduction in the system's capacity to produce electrical energy over time.

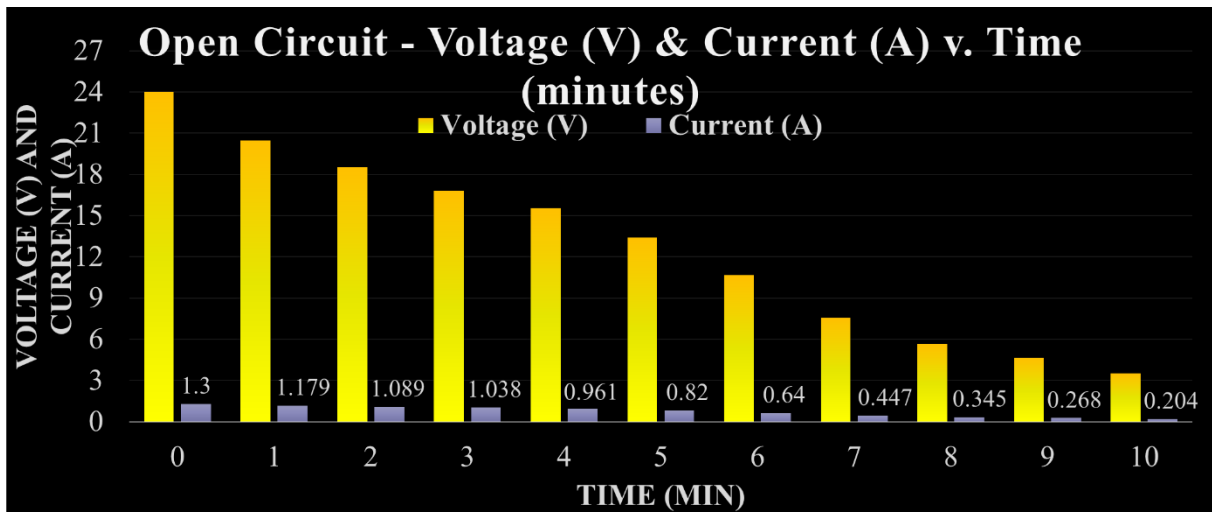


Figure 1: Voltage (V) and Current (A) vs. Time (min) of ThermoElectric Battery

Figure 1 illustrates the change in voltage and current output of the thermoelectric generator (TEG) system over a 10-minute period. The data reveal a decline in both voltage and current over time. Initially, the system exhibits a peak output of 24V and 1.3A, indicating substantial electrical energy generation at the onset. However, as time progresses, both voltage

and current diminish, reaching final measurements of 3V and 0.204A. This decline indicates a reduction in the system's capacity to produce electrical energy over the duration of the experiment. However, the subsequent decline indicates that the system's efficiency decreases as the reaction progresses and the system approaches thermal equilibrium. This trend highlights a limitation of the system: its suitability for short-duration applications.

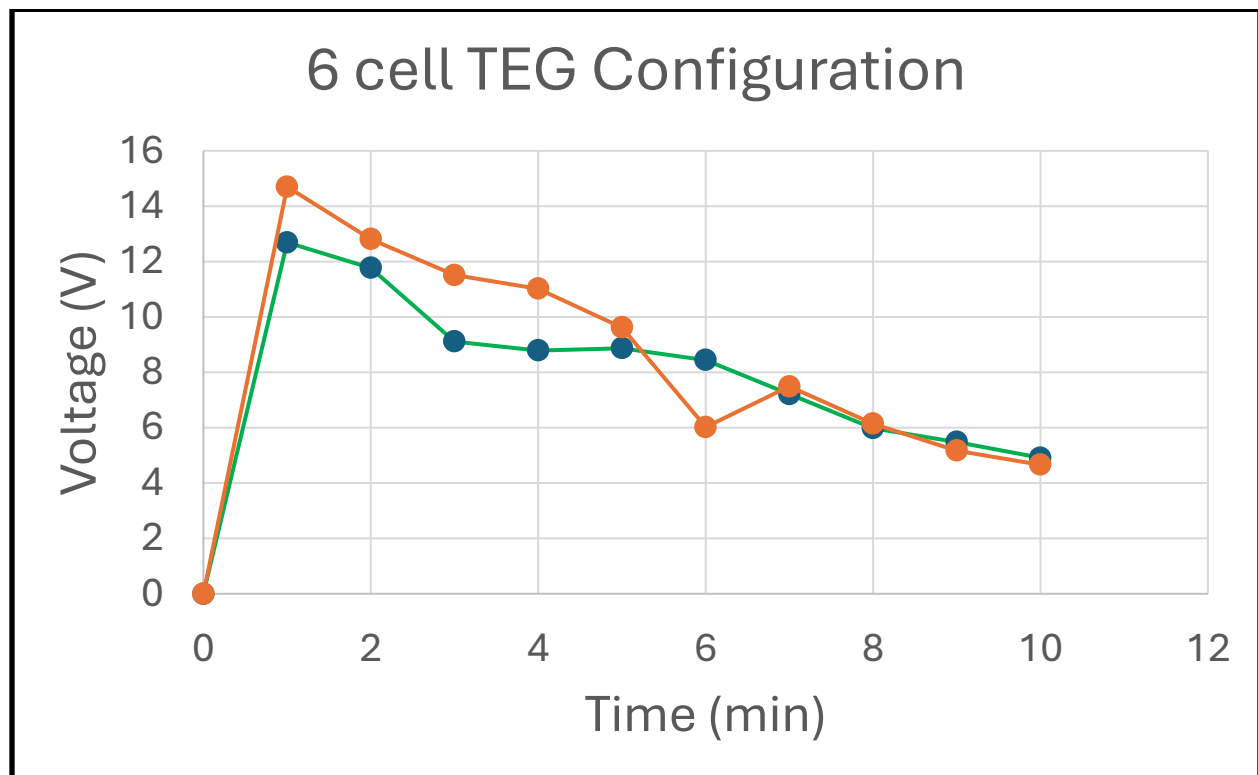


Figure 2: Shows the voltage over time for a 6 TEG configuration cell

While a decrease in voltage and current is expected, the rate and magnitude of the decline may be greater than anticipated. This could be attributed to several factors. Firstly, the rate of the chemical reaction may not be constant, leading to a decrease in heat generation over time. Secondly, there may be significant heat loss to the surroundings, reducing the temperature difference across the TEGs. Finally, the thermal properties of the materials used may affect the

rate at which the temperature gradient dissipates. More data, including more precise temperature measurements and analysis of the reaction kinetics, would be needed to fully explain this rapid decline.

Summary and Conclusion

This report details the design, development, and testing of a chemically powered vehicle for the Chem-E-Car Competition. The vehicle's propulsion system utilized thermoelectric generators (TEGs) to convert thermal energy into electrical power, using an exothermic acid-base reaction (HCl and NaOH) and dry ice. The experimental design focused on optimizing system performance. This was achieved by varying TEG sizes and quantities, and chemical reaction parameters. The final design featured a 3D-printed carbon fiber container with two hot boxes and integrated heat sinks. The system's performance was evaluated by measuring voltage and current output over time.

The experimental results indicate that the TEG system effectively converts chemical energy into electrical power, but its performance is highly dependent on the temperature gradient between the heat source and cold sink. The system exhibits a decline in both voltage and current output over time, with a peak output of 24V and 1.3A decreasing to 3V and 0.204A within 10 minutes. This suggests that the system is best suited for short-duration applications. The 3D-printed carbon fiber container proved to be a durable housing for the system components.

The project successfully demonstrated the feasibility of using TEGs to power a small-scale vehicle using chemical reactions. The final design, incorporating 8 TEGs, an HCl/NaOH reaction, and dry ice, effectively powered a 12V DC motor. However, the decline in power output over time indicates a limitation in its capacity for sustained operation. Further

experimentation and data collection is recommended to enhance sustained power generation and incorporate control mechanisms for improved performance.

Recommendations

To improve the Chem-E car system in future work, there are several important areas to focus on. The main challenge is to get a steadier and longer-lasting power output, because the current setup shows a noticeable drop in power as it operates. One approach is to fine-tune the chemical reaction to ensure it generates heat more consistently. Additionally, adding capacitors to the electrical circuit is a good idea; they can help store energy and provide a more stable power supply to the motor, especially when the car's power demands change quickly. It's also crucial to improve thermal management, specifically by reducing heat loss from the system. This could involve using better insulation or choosing materials with better heat-handling properties, which will help maintain a larger and more stable temperature difference across the thermoelectric generators. Finally, to better understand and optimize the system's performance, it's highly recommended to integrate voltage sensors for real-time power monitoring and temperature sensors to continuously monitor the temperature of the TEGs. Implementing methods for obtaining more accurate temperature measurements at key points in the system will also be beneficial. Addressing these areas should result in a more robust, efficient, and controllable power system for the vehicle.

References

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Appendix

TEG - Model	SP1848-27145	TGMT-19W-4V
Open Circuit Voltage (V)	4.8	8.4
Operating Temperature (°C)	0 to 150	0 to 400
Maximum Temperature (°C)	150	300
Length (mm)	40	56
Width (mm)	40	56
Height (mm)	3.6	5
Weight (g)	30	60

Voltage(V) vs. Run Time (min) of Large and Small TEG

